

Moving Average and Differencing

1. Moving Average

- **What it is:** You take the average of recent values over a fixed period (the “window,” e.g., 30 days).
 - **Why use it:** It smooths out short-term fluctuations (noise) so trends are easier to see.
 - **Limitation:** It does not predict trends or seasonal patterns — sometimes it’s even worse than a naive forecast.
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2. Differencing

- **What it is:** Instead of looking at raw values, you look at the *difference* between a value at time **T** and the value from an earlier time (**T – period**).
 - ✓ **Example:** For daily data with yearly seasonality → $T - 365$ days
 - **Why use it:** This removes trend and seasonality, leaving a stationary series (easier to forecast).
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3. Forecasting the Differenced Series

- Use the moving average on the differenced data.
 - Then, to get the forecast for the *original series*, add back the corresponding past value at $T - \text{period}$.
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4. Noise Problem

- Even after smoothing, adding back past values reintroduces old noise.
 - **Fix:** Apply a moving average to the *past values* before adding them back → results in smoother and more accurate forecasts.
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5. Results

- Naive forecast: MAE ≈ 7.14
 - Moving average on differenced series: MAE ≈ 5.8
 - Smoothed differenced forecast: MAE ≈ 4.5 (very close to the theoretical best ≈ 4).
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➤ Key takeaway:

📌 Simple statistical tricks like *moving average + differencing + smoothing* can come surprisingly close to optimal performance — sometimes making deep learning unnecessary for basic forecasting tasks.
